



ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

Language and Cognitive Development Lab Statistics Workshop

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- explain what's going inside the statistical analyses using *R*.

Why are we doing statistical analysis?

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- We are doing statistics because we want to estimate the effect of variable on the other if it causally makes sense.
- So we do statistics to explain the causal relationship between the variables.





All this absurdity applies to statistics

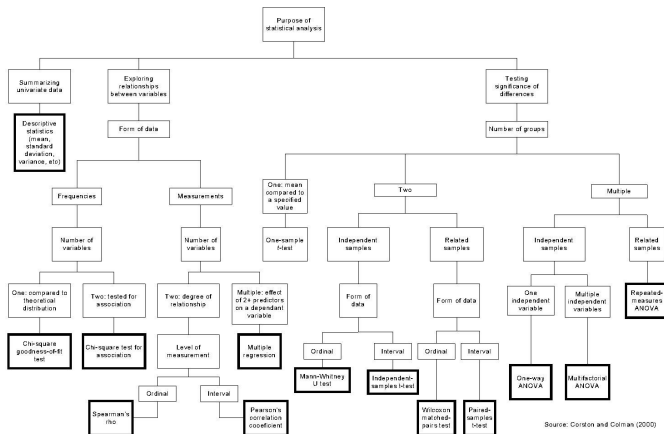


Figure 9. Choosing an appropriate statistical procedure

:)



So what should we do?

- Detect your problem

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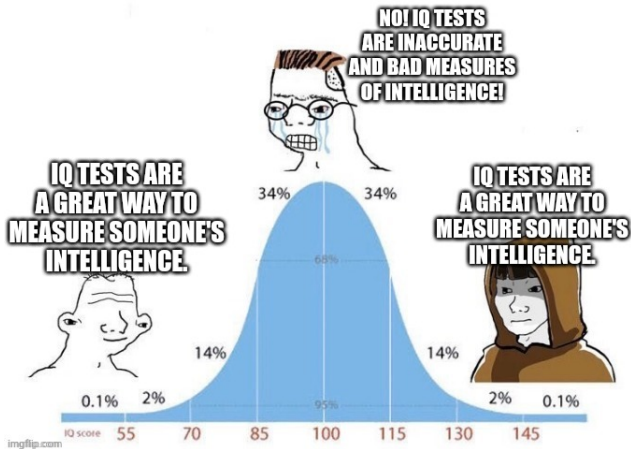
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- Learn what's going on behind the computation

Overburden your brain!



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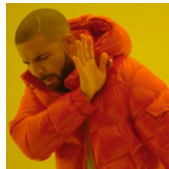
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3. Compute your standard error (SE)
4. Compute your t-value
5. Compute your p-value



Looking
for estimates,
std error,
and t-values



Looking
for
p-values

Overburden your brain!





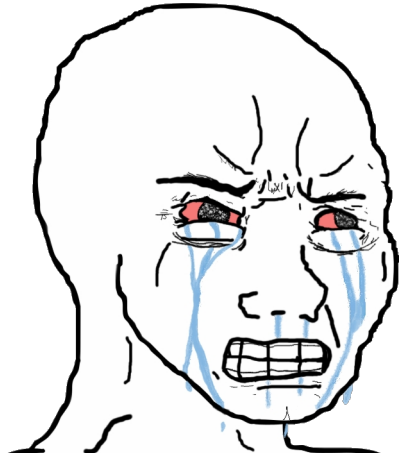
- 1 su bardağı un
- 0,5 çay bardağı Dr. Oetker Mısır Nişastası
- 1 poşet Dr. Oetker Hamur Kabartma Tozu
- 100 g yumuşak margarin
- 1,5 çay bardağı toz şeker
- 1 poşet Dr. Oetker Şekerli Vanilin
- 4 yumurta
- 1 limon kabuğu rendesi



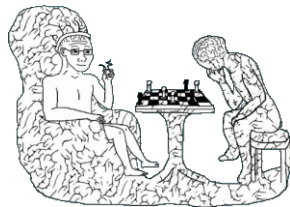
Sometimes :)



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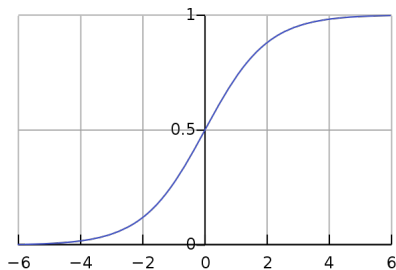
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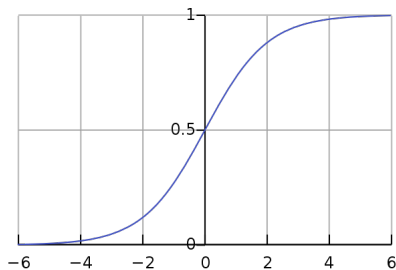
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logistic function

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logistic function

$$\text{logit}(p) \sim \alpha + \beta_1 * x_1 + \beta_2 * x_2$$

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1. Deal with unequal sample sizes (e.g., t-test warns you for that.)
2. Better for repeated measures (no need for aggregation and assuming independence)
3. Shrinkage (learn from the grand mean)
4. More immune to missing data



t-test



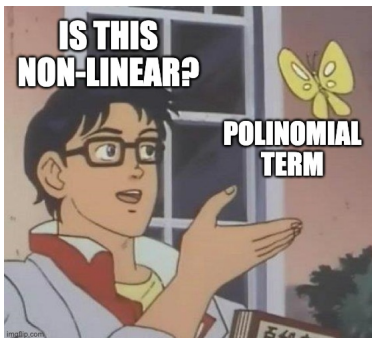
linear mixed effects model with random intercept on aggregated data



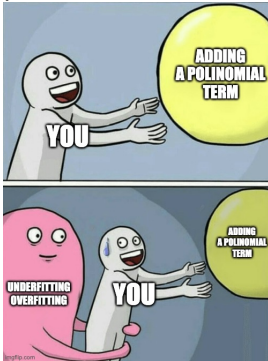
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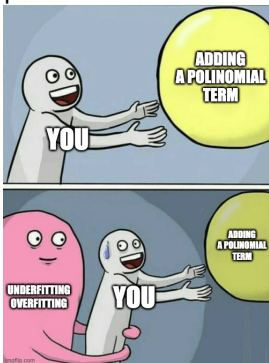
- So far, we have assumed that the relations between the variables are linear.
- Sometimes, these relations may not follow a linear pattern.
- We need to introduce our models such non-linearity.



- Following this might lead to overfitting or underfitting problems.

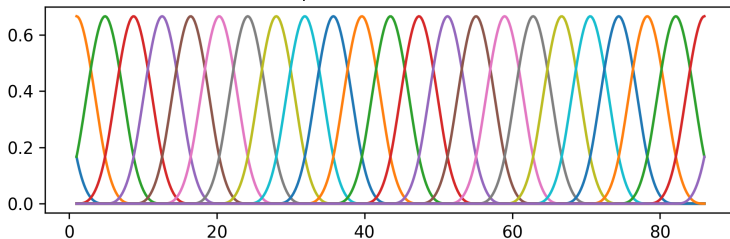


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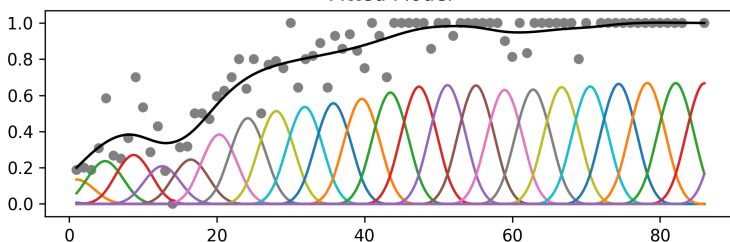


- Solution could be Generalized Additive Model (GAM).

b-Spline Basis Functions



Fitted Model



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- For instance, your model should not predict negative reaction times.
- And your model should not predict values at different scale from the fitted values.
- So, if you feed your model with categories, it should return categories, not some intermediate values.



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When looking ‘hot’ means not feeling cold: Evidence that self-objectification inhibits feelings of being cold

Roxanne N. Felig^{1*} , Jessica A. Jordan¹ , Samantha L. Shepard¹,
Emily P. Courtney¹ , Jamie L. Goldenberg¹  and
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¹Department of Psychology, University of South Florida, Tampa, Florida, USA

²Department of Psychology, Colorado College, Colorado Springs, Colorado, USA

<https://www.youtube.com/watch?v=13bUuooPGxc>

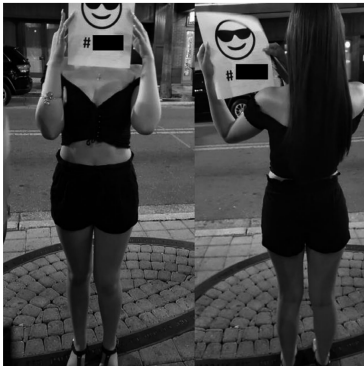


Figure 1. Sample photo of a participant's outfit with redacted ID number. *Note.* The above participant received a score of '7' on the skin exposure variable for meeting the following criteria: open-toed shoes; no jacket; arms, shoulders, legs, chest, and midriff exposed.

Concluding Remarks







Thank You

for your attention.

Do you have any question?

